

OXFORD

INTERNATIONAL
AQA EXAMINATIONS

INTERNATIONAL A-LEVEL CHEMISTRY

(9620) CH03

Report on the examination

January 2024

REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL CHEMISTRY (9620) CH03 JANURARY 2024

This examination of the international specification highlighted some key general points about students' performance that may prove useful.

Students need to:

- learn the meanings of key terms accurately;
- ensure that chemical equations are balanced for both atoms and charge;
- show working out clearly with a clear explanation for each step of their calculation;
- round their numerical answers accurately;
- be familiar with units and be able to use them to calculate quantities,
- be able to draw shapes of molecules and ions in 3D using wedges and dashes,
- understand that in aqueous solution the hexa-aqua ion $[M(H_2O)_6]^{2+}$ or $[M(H_2O)_6]^{3+}$ is present;
- understand that thermodynamics involves enthalpy changes, entropy and Gibbs free-energy changes,
- understand how to construct Hess cycles,
- understand how to give conventional cell representations for electrochemical cells;
- be very familiar with all the practical procedures contained in the specification, and very importantly understand why they carry out the procedures that they do in practical work, rather than knowing just what to do.

QUESTION 01: Period 3 elements and their compounds

- 01.1 The observations were not always well known with some students suggesting that there would be fizzing or a precipitate formed in this reaction. Some students appeared to confuse this reaction with the reaction of sodium with water. In the explanation about why this was a redox reaction, students did not often make reference to oxidation states which meant that only about one third of students scored all the marks on this question
- 01.2 The majority of students were able to name correctly the shape of the PCl_4^+ ion and give the correct bond angle for this tetrahedral structure. However, although the structure of PCl_6^- was often know, students were less familiar with using wedges and dashes to show this in 3D.
- 01.3 This equation was well understood by students, although some students wrote equations that formed P_4O_{10} rather than H_3PO_4
- 01.4 This question was well answered with about three quarters of students correctly identifying the Period 3 element.
- 01.5 Students appeared to find this question challenging with only a low proportion correctly writing two ionic equations to show how aluminium oxide is amphoteric. Students were more confident to write an equation showing how aluminium oxide acts as a base by reacting with H^+ ions, but many struggled to write an accurate equation to show aluminium oxide acting as an acid. Some students gave two correct non-ionic equations showing how aluminium oxide is amphoteric.

QUESTION 02: Thermodynamics

- 02.1 This was well answered by the majority of students. However, some students described the observations of the reaction of sodium with water rather than the reaction of sodium with oxygen.

02.2 Students generally were able to accurately calculate the standard entropy of $\text{O}_2(\text{g})$, although some students did not recognise the need to take the stoichiometric coefficients from the equation into account. Students appeared to be less confident when commenting on the difference in the entropy values for O_2 and Na and did not always understand or articulate the link between the entropy value and the disorder of the substance.

02.3 The majority of students were able to complete correctly the Born-Haber cycle with very few students not scoring at least one mark. Common errors included missing or incorrect state symbols, or missing electrons where these were needed.

02.4 Students were generally able to complete the Born-Haber cycle correctly. However, students did not always appreciate that the cycle involved two sodium atoms and so the enthalpy of atomisation of sodium and the first ionisation energy for $\text{Na}(\text{g})$ values should both be doubled. Working out was not always clear and some students just added all the numbers together rather than applying Hess' Law to the cycle.

02.5 The majority of students were able to calculate a value for the enthalpy of hydration of $\text{Cl}^-(\text{g})$. However, fewer students were able to complete the Hess' cycle correctly. Suitable species (with appropriate state symbols) should have been shown along with the values for the enthalpy of solution and the enthalpy of lattice dissociation.

02.6 Students did not appreciate that Na_2O does not dissolve in water but instead reacts with water to form $\text{Na}^+(\text{aq})$ and $\text{OH}^-(\text{aq})$.

QUESTION 03: The reaction between ethanedioate ions and manganate(VII) ions

03.1 This question was well answered by the majority of students. Students that failed to score on this question did not appreciate that there would be repulsion between the manganate(VII) and ethanedioate ions.

03.2 Equations that show how Mn^{2+} ion can act as a catalyst for this reaction were not well known and less than one quarter of students scored both marks.

03.3 This calculation was well completed with many students scoring all the marks. Common errors in calculations included using non-concordant titres in calculating the mean titre, using an incorrect reacting ratio. Some students did not show the M_r of the solid metal ethanedioate but instead gave the M_r of the metal ion.

03.4 This was reasonably well answered with many students accurately describing the colour change at the end point. However some students did not describe a change in colour and only gave the final colour of the solution. The reason for the final pink colour was sometimes confused as being due to the presence of $\text{Mn}^{2+}(\text{aq})$ ions rather than an excess of MnO_4^- ions.

QUESTION 04: Bidentate and Multidentate ligands

04.1 A good number of students were able to show correctly how the acac ion is able to act as a bidentate ligand but a large number of students did not attempt this question.

04.2 A large majority of students were able to give the correct oxidation state of cobalt in the $\text{Co}(\text{C}_5\text{H}_7\text{O}_2)_3$ complex and correctly identify the type of isomerism shown by this complex.

04.3 This question was found by students to be challenging with only a few students scoring all six marks. When explaining, in terms of thermodynamics, students need to discuss the enthalpy change,

entropy change and the Gibbs free-energy change for the ligand substitution reaction. A large number of students only focussed on the entropy change. A common misconception was for students to think that the strength of the coordinate bonds in the EDTA complex was higher than in the hexa-aqua complex.

QUESTION 05: Reactions of Aqueous Solutions

05.1 About half of the students provided a correct equation for the formation of the deep-blue solution $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$. However, some students gave this complex as a neutral complex. It was very pleasing to see that the vast majority of students were able to identify that solution **P** contained Cu^{2+} ions.

05.2 This was well answered by the vast majority of students. However, the expression relating the energy gap (ΔE) to wavelength was not always known by students. Wavelengths were not always converted to meters for use in the expression.

05.3 This was usually well answered by students although a good number did not appreciate that this was a ligand substitution reaction. Equations, when given, were usually correct although a few students did not balance the equations for both atoms and charge.

05.4 The majority of students were able to correctly identify a reagent and expected observations that show the presence of sulfate(VI) ions. Some students chose barium hydroxide which would also have formed an additional precipitate through the reaction of the hydroxide ions with the metal ion present in solution **P**.

5.5 Most students were able to correctly identify one of the precipitates but often did not realise that the green precipitate formed in Test 2 would have been FeCO_3 . A few students felt that the green precipitate formed in Test 1 was a complex where ammonia ligands had substituted onto the Fe^{2+} ion.

5.6 This equation was often not correctly balanced. A good number of students wrote equations to produce $\text{Fe}_2(\text{CO}_3)_3$ rather than $[\text{Fe}(\text{H}_2\text{O})_3(\text{OH})_3]$

QUESTION 06: Fuel Cells

06.1 This question was found to be demanding with only about a third of students scoring all three marks. Half-equations were often not given for the electrode reactions and students did not always use the information in the question stem to help them construct these. It was pleasing to see that some students realised that the overall reaction that occurs in the ethanol fuel cell was the same as that of the combustion of ethanol so scored this mark even if the electrode half equations were not correct.

06.2 This was well answered with the vast majority of students correctly calculating the electrode potential for the negative electrode.

06.3 Students found this demanding and often did not appreciate the need for a constant supply of ethanol (fuel) to maintain constant EMF.

06.4 Very few students scored both marks and were able to give a correct conventional cell representation. Answers often did not show phase boundaries or Pt electrodes. Some students gave an equation for the cell reaction rather than the cell representation.

06.5 This was not well answered. Students did not often consider the issues surrounding storage and transport of hydrogen in their answers.

QUESTION 07: Acids and Bases

07.1 This question was well completed with the majority of students being able to define pH.

07.2 Students found this question challenging with about half scoring both marks. A variety of creditworthy approaches were taken. Students did not always make correct use of the density of the concentrated acid solution in their calculations.

07.3 This was found to be challenging with about half of the students scoring all the marks. Most students were able to work out concentration of H^+ ions from the desired pH. Students often did not then use this desired concentration of H^+ to work out the number of moles needed in 4.00 dm^3 in order to then determine the volume of the more concentrated acid needed.

07.4 This was a demanding concentration which required students to realise that the barium hydroxide releases 2 moles of hydroxide ions per mole of solid when dissolved in water. Students also often did not convert their moles of hydroxide ions to a concentration when determining the concentration of H^+ ions in the solution.

07.5 This was not well answered. Students did not commonly make the link between pH and solubility of group II metal hydroxides. A common misconception was discussion of covalent character to help explain the difference in pH

07.6 This was well answered with the majority of students being able to determine the pK_a value from the acid dissociation constant.

07.7 About half of the students scored all 5 marks in this challenging calculation. However, some students did not appreciate that when sodium hydroxide solution is added to excess weak acid, then an acidic buffer is formed. Some students therefore treated this mixture as being a pure weak acid and so assumed in their calculation that the $[\text{A}^-]$ and $[\text{H}^+]$ were the same. Students needed to work out the amount, in moles, of excess acid by subtraction and the amount, in moles, of salt formed and use these values to determine the $[\text{H}^+]$ in the mixture.

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